Project Design Phase-I
Solution Architecture

Date	19 September 2022
Team ID	699ee179f292113d819d8e0e
Project Name	Malaria Detection Using Deep Learning (CNN)
Maximum Marks	4 Marks

Solution Architecture:

The proposed malaria detection system uses deep learning and web technologies to automatically classify microscopic blood smear images. The system architecture consists of multiple layers that work together to process medical images and generate diagnostic predictions.

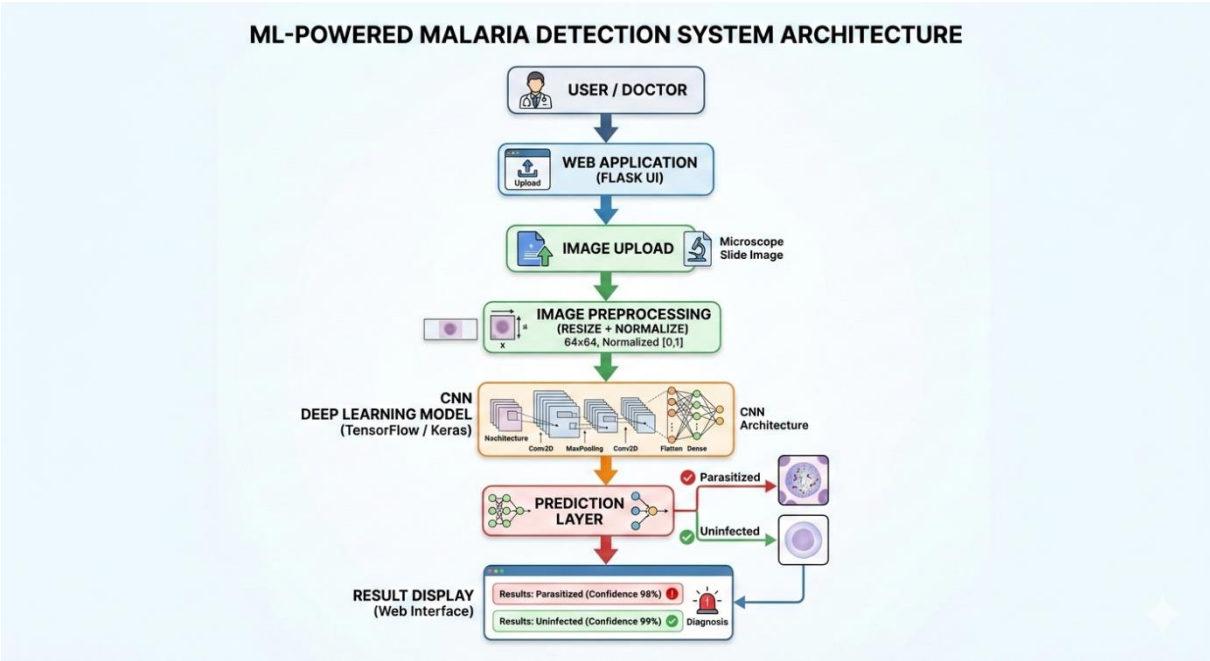
The architecture begins with the user interface, where healthcare professionals upload microscopic blood smear images through a web application. The application is developed using the Flask framework and provides an easy interface for image submission.

After an image is uploaded, the image preprocessing module resizes the image to 224×224 pixels and normalizes pixel values. This preprocessing ensures that the input image is compatible with the trained deep learning model.

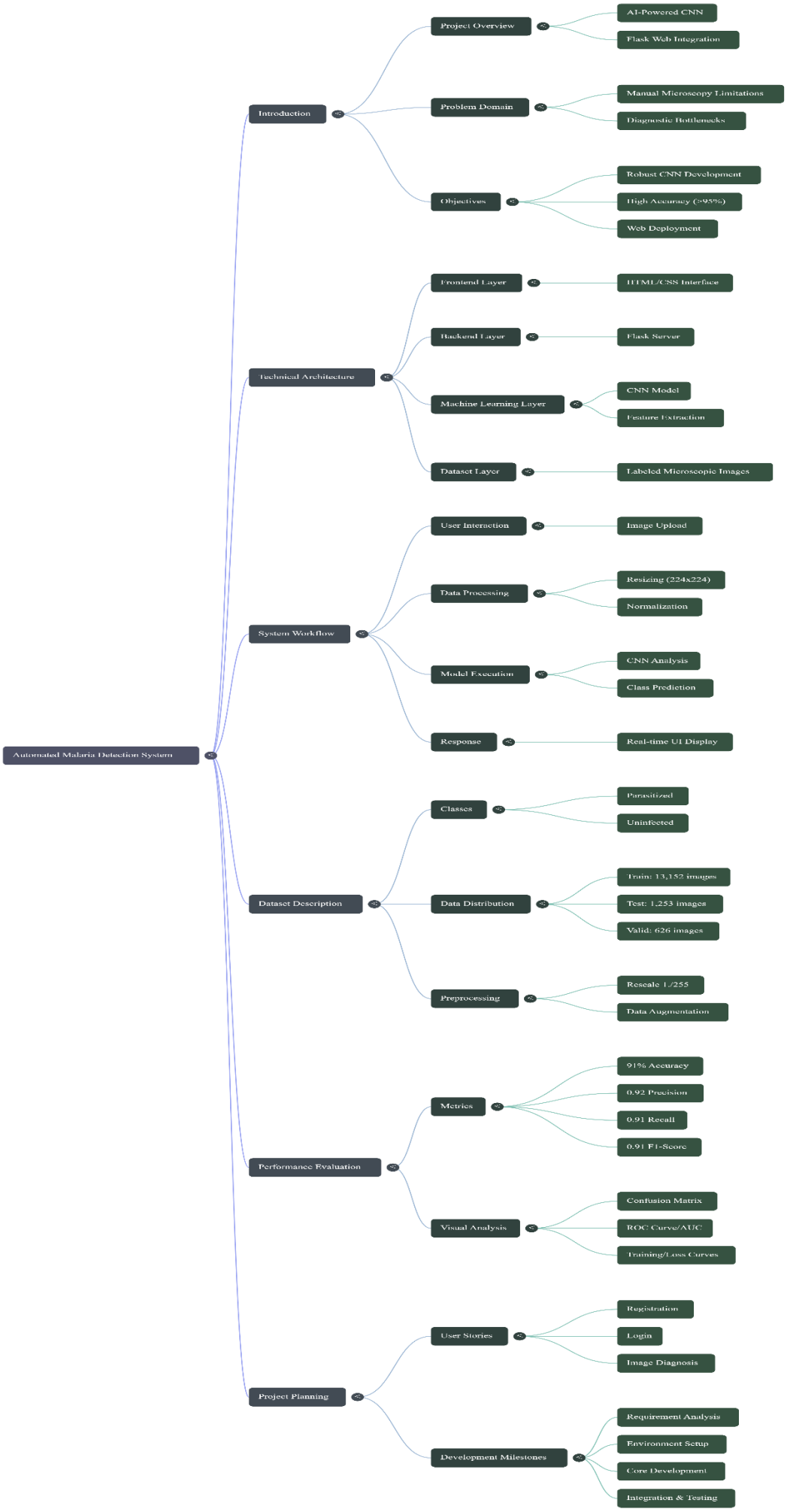
The processed image is then passed to the Convolutional Neural Network (CNN) model, which has been trained using a dataset of parasitized and uninfected blood cell images. The CNN extracts spatial features from the image and classifies it into two categories: infected or healthy.

The prediction module analyzes the model output and determines whether the uploaded cell image is parasitized or uninfected. The prediction results are then returned to the web application interface.

Finally, the system displays the diagnostic result to the user, helping doctors and laboratory technicians quickly identify malaria infections.



Example - Solution Architecture Diagram:



Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/